

8.7 Classwork: Function Composition

1. [Maximum mark: 6]

The functions f and g are defined in the following table.

x	-2	0	1	3	5
$f(x)$	3	1	5	-2	0
$g(x)$	1	-2	0	5	3

- (a) Write down the value of $f(1)$. [1]

- (b) Find the value of $(g \circ f)(1)$. [2]

- (c) Find the value of $(f \circ g)(-2)$. [2]

- (d) Find the value of $f^{-1}(-2)$. [1]

2. [Maximum mark: 7]

Let $f(x) = 2x - 1$ and $g(x) = x^2 + 3$.

(a) Find $(f \circ g)(x)$. [2]

(b) Find $(g \circ f)(x)$, giving your answer in expanded form. [2]

(c) Evaluate $(g \circ f)(-1)$. [1]

(d) Solve the equation $(f \circ g)(x) = 13$. [2]

3. [Maximum mark: 7]

Let $f(x) = \frac{1}{x-2}$ and $g(x) = x + 5$.

(a) Find $(f \circ g)(x)$, simplifying your answer. [2]

(b) Write down the value of x that must be excluded from the domain of $f \circ g$. [1]

(c) Find $(g \circ f)(x)$, giving your answer as a single fraction. [2]

(d) Find $(f \circ f)(x)$, simplifying your answer. [2]

4. [Maximum mark: 6]

The function f is defined by $f(x) = 3x - 4$, and $(f \circ g)(x) = 6x + 2$.

(a) Find an expression for $g(x)$.

[3]

(b) Find $g^{-1}(x)$.

[3]